Senior Cloud/DevOps Engineer – Technology People – 15/8/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cloud/DevOps Engineer (Hybrid) /Integration Solution Architect /Software Engineer/Senior Data Engineer /Java/Salesforce/Mulesoft Developer /Python AI/ML/DL/BI/.NET/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Cloud/DevOps Engineer/Snr Data Engineer/Python/Go/AI/ML/DL/Salesforce/SharePoint/D365 Developer
Responsibilities:

- Designed and implemented a secure hybrid cloud architecture integrating AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted site-to-site connections using Cloud VPN Gateways across all platforms. Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python (Boto3) with CloudFormation, Terraform, or Ansible. Utilized ECS Fargate for efficient and scalable container orchestration.
- Architect and implement scalable AWS infrastructures using Python (Boto3) & CloudFormation to meet business objectives.
- Ensured robust security and compliance of AWS environments through the effective use of CloudTrail and AWS Config.
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., `dfx force:source:pull`) from Salesforce orgs.
- Built secure and scalable AWS Landing Zones using Infrastructure as Code (IaC) with AWS CloudFormation templates or Terraform. Leveraged AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch.
- Design and lead the migration of complex applications from Rancher/Docker on EC2 to ECS Fargate, optimizing performance and cost efficiency. Used CloudWatch Contributor Insights to analyze High-Cardinality Data & Traffic Trends.
- Built and managed ECS clusters and task definitions, ensuring seamless application deployment and management.
- Implemented robust CI/CD pipelines for automated builds, testing, and deployment to ECS Fargate.
- Established comprehensive monitoring and optimization strategies to maintain application health and performance.
- Successfully migrated critical applications with minimal downtime and disruption to business operations.
- **CloudWatch Observability Expertise:** Implemented comprehensive CloudWatch observability solutions, including metrics, logs, and dashboards. Leveraged CloudWatch Insights for advanced log analysis and troubleshooting.
- Implemented custom CloudWatch metrics using Boto3 to track critical application performance indicators, enabling data-driven decision-making and proactive issue resolution: cloudwatch = boto3.client('cloudwatch')
- Python or Go programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Automation**:Utilized AWS CodeBuild, Git Actions, GitLab CI, or Jenkins for automated build, test, & deployment processes. Used Ansible, Chef, or Puppet for configuration management & infrastructure provisioning in cloud-agnostic environments.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
    - **Azure: Azure DevOps**
    - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro.
- Combining Lean's optimizing processes and eliminating waste and Agile's flexibility and responsiveness to drive significant improvements in efficiency, quality, and customer satisfaction.

- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, RNN, ANN, GAN) based on dataset type, size, and desired result interpretability.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior Cloud/DevOps Engineer/Senior Data Engineer/Java/Python /Go/AI/ML/BI/SP Developer /EDW
Responsibilities:

- **Built and implemented Medallion Architecture with Data Lake in GCP:**
- **Batch Layer:** Utilized Google Cloud Dataproc, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, & unrefined data.
- **Serving Layer:** Implemented BigQuery with its built-in Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer:** Leveraged Cloud Pub/Sub as the unified service layer for real-time data ingestion and BigQuery for querying the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
  Utilized BigQuery for data querying, loading, exporting to Pandas DataFrame, and executing parameterized queries within the Python programming environment. query_job = client.query(query, job_config=job_config)
- Designed and developed Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/ J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
- Full-stack engineer skilled in React/TypeScript for frontend development, Java/APIs for backend development, and AWS cloud. Proficient in integration tools like MuleSoft and J2EE to ensure seamless data flow using RESTful web services.
- Combined Lean and Agile methodologies to optimize processes, eliminate waste, and drive significant improvements in efficiency, quality, and customer satisfaction.

- Collaborated with business analysts and product owners to translate data requirements into actionable solutions, driving data-driven decision-making. Utilized Talend for ETL/ELT processes, achieving streamlined data integration.
- Expand capabilities across the entire data pipeline lifecycle, encompassing the building, running, testing, and operating of data pipelines that ingest, filter, sort, join, and/or transform datasets on both GCP and SAP Data Services by utilization of Cloud Build for pipeline automation, GitOps for infrastructure-as-code management, and Argo CD for continuous delivery.
  - **Automation and CI/CD**: Leveraged industry-standard tools like Helm and Kubespray to streamline & customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
  - Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.
  - Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.
  - Developed and deployed ML models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision): model = torch.hub.load('pytorch/vision:v0.6.0', 'mobilenet_v2', pretrained=True)
  - Expertise in implementing and maintaining a robust suite of cloud security controls, including Identity and Access Management (IAM), data encryption, network security, vulnerability management, and data loss prevention (DLP). Ensured compliance with industry standards and regulations such as CIS, NIST, GDPR, and HIPAA.
  - Design, develop, and maintain a robust ETL pipeline/API orchestration using either:
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
  - Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
  - **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
  - Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
  - Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
  - Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
  - Proficient in Linux system administration, including kernel optimization (sysctl, vmstat), performance tuning (iostat, sar), and security hardening (iptables, fail2ban).

- Developed and deployed ML models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision: model = models.detection.fasterrcnn_resnet50_fpn(pretrained=True)
- Applied technical expertise in monitoring, observability, performance management, risk management, troubleshooting, and service recoverability. Executed and reported on non-functional testing, contributing to system improvements.

- Migrated on-premise applications to the Google Cloud Platform (GCP) cloud using Velostrata, Cloud Migrate for Compute Engine, Migrate for Anthos, and CloudEndure Migration to GKE environment.
  - Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Worked with enterprise CI/CD capabilities and tools such as GoCD, GitLab, Code Build, and Cloud Formation to provide guidance on best practices and help enable major programs to deliver business value.
- Deploy and manage microservices-based applications using Docker CRE and Kubernetes on GCP GKE clusters.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Cloud/DevOps Engineer/Integration Solution Architect/Java/.NET/Python/Mulesoft/SP/K2 Developer
Responsibilities:

- Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications. Build and maintain all non-production and production environments.
- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, Dynamics 365 CRM, and Nintet K2 Five workflows. Additionally, the platform integrates with data visualization tools like Power BI and Tableau for further data analysis.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Automated Salesforce Development Workflow:
- Utilized Salesforce DX tools (`dfx force:source:pull`) to retrieve and manage code (metadata) from development orgs.
- Implemented unit testing frameworks like Jest to ensure code quality on retrieved Salesforce code.
- Configured CI/CD pipelines to automate code retrieval, testing, & deployment to staging orgs using `dfx force:source:deploy`. (Optional: additional deployment steps like code analysis and MuleSoft application deployments if applicable)
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., `dfx force:source:pull`) from Salesforce orgs.
- **Automated Deployments with SFDX:** Minimized manual errors and accelerated releases.
- **Enhanced Code Quality**: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.
- **MuleSoft Integration**: Built seamless Salesforce-MuleSoft integrations within CI/CD for end-to-end automation.
- Designed and maintained robust CI/CD pipelines using platform-specific tools:
  1. **Azure: Azure DevOps**
  2. **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
  3. **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- **Automated Kubernetes Deployments:** Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.
- Proven ability to design & implement highly scalable, resilient, & cost-effective cloud-native applications on the Azure platform, leveraging technologies such as Azure Service Bus, Azure Event Hubs, Azure Container Service (ACS), and Azure DevOps.
- **Secured API Access for Student Enrollment**: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of students' data.
- **CI/CD and DevSecOps Automation:** Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or even on-premises environment.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Cloud/DevOps Engineer/Data Engineer /Software Test Engineer/Linux Administrator
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed Amazon Redshift with its built-in data processing capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- Developed and maintained automation scripts using Python, Go, Bash to streamline development & operational workflows.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.
- Collaborate with development and operations teams to streamline the deployment process of our Python Web applications.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within AWS cloud architecture.
- **Security**: Secured secrets with HashiCorp Vault or AWS Secrets Manager and implemented network segmentation and firewalls. Integrated security scanning tools throughout development.
- **Leverage AWS services for cloud-native design patterns**:
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- **Optimized CI/CD pipelines with DevOps tools**: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon Elastic Container Registry (ECR) and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Successfully leveraged ITIL 4.0 frameworks, including Incident Management, Problem Management, and Continual Improvement, to optimize incident response processes and drive service excellence.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Linux Server Provisioning: Utilize Ansible playbooks (e.g., ec2_instance.yml, configure_router.yml, network_config.yml, install_packages.yml, deploy_app.yml) for creating EC2 instances, configuring networks, installing packages, and deploying applications.
- Windows Server Configuration: Employ Chef, initiated with knife bootstrap windows_server_ip chef-server_url, for managing Active Directory, group policies, and software installations.
- Database Provisioning: Leverage Puppet to automate database creation (e.g., mysql_database { 'mydatabase': ensure => present, }) and user management.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Cloud/DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using AWS API Gateway
- Deploy and maintain web applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).

- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Sora, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Phi 3, Copilot, Google GEMINI, Metaverse Llama 3.1, Mistral SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0

- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

DevOps Engineer – GBST – 13/8/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer (Hybrid) /Integration Solution Architect /Software Engineer/Senior Data Engineer /Java/Salesforce/Mulesoft Developer /Python AI/ML/DL/BI/.NET/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DevOps Engineer /Senior Data Engineer /Python/Go/AI/ML/DL/Salesforce/SharePoint/D365 Developer
Responsibilities:

- Designed and implemented a secure hybrid cloud architecture integrating AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted site-to-site connections using Cloud VPN Gateways across all platforms. Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python (Boto3) with CloudFormation, Terraform, or Ansible. Utilized ECS Fargate for efficient and scalable container orchestration.
- Architect and implement scalable AWS infrastructures using Python (Boto3) & CloudFormation to meet business objectives.
- Ensured robust security and compliance of AWS environments through the effective use of CloudTrail and AWS Config.
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., `dfx force:source:pull`) from Salesforce orgs.
- Built secure and scalable AWS Landing Zones using Infrastructure as Code (IaC) with AWS CloudFormation templates or Terraform. Leveraged AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch.
- Design and lead the migration of complex applications from Rancher/Docker on EC2 to ECS Fargate, optimizing performance and cost efficiency. Used CloudWatch Contributor Insights to analyze High-Cardinality Data & Traffic Trends.
- Built and managed ECS clusters and task definitions, ensuring seamless application deployment and management.
- Implemented robust CI/CD pipelines for automated builds, testing, and deployment to ECS Fargate.
- Established comprehensive monitoring and optimization strategies to maintain application health and performance.
- Successfully migrated critical applications with minimal downtime and disruption to business operations.
- **CloudWatch Observability Expertise:** Implemented comprehensive CloudWatch observability solutions, including metrics, logs, and dashboards. Leveraged CloudWatch Insights for advanced log analysis and troubleshooting.
- Configured CloudWatch alarms to proactively monitor system health and performance.
- Optimized CloudWatch costs through efficient data retention and resource management.
- Implemented custom CloudWatch metrics using Boto3 to track critical application performance indicators, enabling data-driven decision-making and proactive issue resolution: cloudwatch = boto3.client('cloudwatch')
- Collaborated with development teams to identify and instrument key performance indicators (KPIs).
- Python or GoLang programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Automated Builds**: Utilize AWS CodeBuild for automated build, test, and deployment processes.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Technical Proficiency:** Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
  - **Azure: Azure DevOps**
  - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
  - **GCP: Cloud Build/Cloud Deploy**

- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior DevOps Engineer/Senior Data Engineer/Mulesoft/Java/Python/AI/ML/DL/BI/SP/D365 Developer
Responsibilities:

- Developed and deployed a secure hybrid cloud infrastructure by integrating AWS VPCs, GCP Cloud VPCs, and Azure VNets. Implemented encrypted site-to-site connections across all platforms using Cloud VPN Gateways. Automated the provisioning and configuration of network resources with Infrastructure as Code (IaC) principles, utilizing Python (Boto3) alongside CloudFormation, Terraform, or Ansible. Employed ECS Fargate for scalable and efficient container orchestration.
- Designed and developed Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
- Built secure & scalable GCP Landing Zones using IaC with Terraform or Cloud Build. Utilized GCP Cloud IAM for centralized governance and management of multiple GCP projects, enabling efficient parallel processing for various projects.
- Led complex migration projects, encompassing application assessment, dependency mapping, and image optimization.
- Used CloudWatch Contributor Insights for in-depth analyse high-cardinality data & traffic patterns to optimize performance.
- Executed seamless migrations from Rancher/Docker to AWS ECS Fargate, improving operational efficiency.
- Developed and maintained efficient CI/CD pipelines using AWS CodePipeline, CodeBuild, and CodeDeploy for rapid application deployment. Provided technical guidance and support to customers, ensuring smooth operations.
- Implemented robust monitoring and logging solutions with CloudWatch to optimize performance and troubleshoot issues.
- Created CloudWatch Dashboards and visualized metrics and logs with customizable CloudWatch dashboards.
- Used widgets to display graphs, tables, and maps.
- Set up CloudWatch Alarms based on metric conditions, log patterns, or by using Python programming with the Boto3 CloudWatch client: `cloudwatch_logs = boto3.client('logs')`
- Configure notification actions, such as sending emails or SNS messages.
- Developed and executed comprehensive migration plans, including application assessment, dependency mapping, and image optimization. Utilized Amazon CloudWatch Contributor Insights to Analyze Traffic Trends
- Proficient in designing and implementing robust observability solutions using CloudWatch, including custom metric creation and advanced log analysis with CloudWatch Insights. Leveraged Python (Boto3) for efficient metric publication.
- Developed Mulesoft Anypoint APIs and Dashboards to provide real-time insights and monitoring of trading activities.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- Developed and maintained automation scripts using Python, Go, Bash to streamline development & operational workflows.
- Implemented automated security scans (SAST, DAST) into the GitLab CI pipeline, enforced granular access controls using roles, and managed secrets securely. Regular audits ensured compliance with industry standards like PCI-DSS or SOC 2.
- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
    - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: `template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']`
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.

- **CI/CD for Integration Solutions:** Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (MuleSoft, Cloud Functions, or Hybrid).
- **Data Transformation Expertise:** Utilized MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations and DBT for standardization and basic transformations.
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
- **CI/CD and DevSecOps Automation:** Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.
- **Scalable Infrastructure**: Integrate with RDS for databases, S3 for storage & Bitbucket repositories for code management.
- **AI-Powered Pattern Recognition:** Used Artificial Neural Networks to model complex patterns and forecast potential issues.
- **High-Performance Kubernetes Cluster:** Built a high-performance Kubernetes cluster on Nvidia DGX-1 systems with pre-installed DGX OS for running containerized AI, HPC, and industry-specific workloads.
- Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation & management of deployed ML models in production env.
- Delivered optimal ML solutions either using Python, R with SQL 2016 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer /Integration Solution Architect/Java/.NET/Python/Mulesoft/SP/K2 Developer
Responsibilities:

- Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications. Build and maintain all non-production and production environments.
- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, Dynamics 365 CRM, and Nintet K2 Five workflows. Additionally, the platform integrates with data visualization tools like Power BI and Tableau for further data analysis.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Automated Salesforce Development Workflow:
- Utilized Salesforce DX tools (`dfx force:source:pull`) to retrieve and manage code (metadata) from development orgs.
- Implemented unit testing frameworks like Jest to ensure code quality on retrieved Salesforce code.
- Configured CI/CD pipelines to automate code retrieval, testing, & deployment to staging orgs using `dfx force:source:deploy`. (Optional: additional deployment steps like code analysis and MuleSoft application deployments if applicable)
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., `dfx force:source:pull`) from Salesforce orgs.
- **Automated Deployments with SFDX:** Minimized manual errors and accelerated releases.
- **Enhanced Code Quality**: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.
- **Streamlined Change Management**: Utilized Salesforce change sets (or a version control system like Git, GitLab, Azure Repos, or AWS CodeCommit) within the CICD flow for managing code changes.
- **MuleSoft Integration**: Built seamless Salesforce-MuleSoft integrations within CI/CD for end-to-end automation.
- Designed and maintained robust CI/CD pipelines using platform-specific tools:
    1. **Azure: Azure DevOps**
    2. **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    3. **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- **Automated Kubernetes Deployments:** Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.
- Design and implement secure & scalable CI/CD pipelines using AWS CodeBuild, RDS & S3 with Bitbucket repositories.
- Production) and managed resource allocation.
- **Secured API Access for Student Enrollment**: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of students' data.

- **CI/CD and DevSecOps Automation (AWS)**: Built and maintained CI/CD pipelines for various applications, employing DevSecOps practices using industry-standard tools (CodeBuild/GitHub) & following best practices for secure development
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or even on-premises environment.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: DevOps Engineer/Data Engineer /Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code (IaC) tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed Amazon Redshift with its built-in data processing capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- Developed and maintained automation scripts using Python, Go, Bash to streamline development & operational workflows.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.
- Collaborate with development and operations teams to streamline the deployment process of our Python Web applications.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within AWS cloud architecture.
- **Security**: Secured secrets with HashiCorp Vault or AWS Secrets Manager and implemented network segmentation and firewalls. Integrated security scanning tools throughout development.
- Troubleshoot and resolve issues related to deployment, configuration, and performance.
- **Leverage AWS services for cloud-native design patterns**:
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- **Optimized CI/CD pipelines with DevOps tools**: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon Elastic Container Registry (ECR) and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**

Government Council: AWS DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using AWS API Gateway
- Deploy and maintain web applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Aug.2009 - Oct. 2011 Manulife Financial Corporation (MLJ), Tokyo
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 ING Life Insurance Company, LTD, Tokyo
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 Microsoft Japan, Tokyo
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Sora, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Phi 3, Copilot, Google GEMINI, Metaverse Llama 3.1, Mistral SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior AWS DevOps Engineer – Nauge Technology Group – 11/8/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer (Hybrid) /Integration Solution Architect /Software Engineer/Senior Data Engineer /Java/Salesforce/Mulesoft Developer /Python AI/ML/DL/BI/.NET/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior AWS DevOps Engineer /Python/AI/ML/DL/SP/D365 CRM Developer
Responsibilities:

- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code (IaC) tools like Terraform or Ansible for automated provisioning and configuration of networks, and utilized ECS Fargate for efficient and scalable container orchestration.
- Architect and implement scalable AWS infrastructures using Python (Boto3) & CloudFormation to meet business objectives.
- Ensured robust security and compliance of AWS environments through the effective use of CloudTrail and AWS Config.
- Collaborated closely with stakeholders to translate business requirements into secure, cost-effective AWS solutions.
- Built secure and scalable AWS Landing Zones using Infrastructure as Code (IaC) with AWS CloudFormation templates or Terraform. Leveraged AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch.
- Design and lead the migration of complex applications from Rancher/Docker on EC2 to ECS Fargate, optimizing performance and cost efficiency. Used CloudWatch Contributor Insights to analyze High-Cardinality Data & Traffic Trends.
- Built and managed ECS clusters and task definitions, ensuring seamless application deployment and management.
- Implemented robust CI/CD pipelines for automated builds, testing, and deployment to ECS Fargate.
- Established comprehensive monitoring and optimization strategies to maintain application health and performance.
- Successfully migrated critical applications with minimal downtime and disruption to business operations.
- **CloudWatch Observability Expertise:** Implemented comprehensive CloudWatch observability solutions, including metrics, logs, and dashboards. Leveraged CloudWatch Insights for advanced log analysis and troubleshooting.
- Configured CloudWatch alarms to proactively monitor system health and performance.
- Optimized CloudWatch costs through efficient data retention and resource management.
- Implemented custom CloudWatch metrics using Boto3 to track critical application performance indicators, enabling data-driven decision-making and proactive issue resolution: cloudwatch = boto3.client('cloudwatch')
- Collaborated with development teams to identify and instrument key performance indicators (KPIs).
- Python or GoLang programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …

- **Automated Builds**: Utilize AWS CodeBuild for automated build, test, and deployment processes.
- **Scalable Infrastructure**: Integrate with RDS for databases, S3 for storage, & Bitbucket repositories for code management.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Optimized CI/CD pipelines with DevOps tools**: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
- Implement effective branching strategies utilizing Gitflow for seamless deployments and continuous application monitoring.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
    - **Azure: Azure DevOps**
    - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior AWD SevOps Engineer/Snr Data Engineer/Mulesoft/Java/Python/AI/ML/DL/BI/SP/D365 Developer
Responsibilities:

- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code (IaC) tools like Terraform or Ansible for automated provisioning and configuration of networks, including ECS Fargate environments, ensuring consistency and efficiency.
- Designed and developed Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
- Built secure & scalable GCP Landing Zones using IaC with Terraform or Cloud Build. Utilized GCP Cloud IAM for centralized governance and management of multiple GCP projects, enabling efficient parallel processing for various projects.
- Led complex migration projects, encompassing application assessment, dependency mapping, and image optimization.
- Used CloudWatch Contributor Insights for in-depth analyse high-cardinality data & traffic patterns to optimize performance.
- Executed seamless migrations from Rancher/Docker to AWS ECS Fargate, improving operational efficiency.
- Developed and maintained efficient CI/CD pipelines using AWS CodePipeline, CodeBuild, and CodeDeploy for rapid application deployment. Provided technical guidance and support to customers, ensuring smooth operations.
- Implemented robust monitoring and logging solutions with CloudWatch to optimize performance and troubleshoot issues.
- Created CloudWatch Dashboards and visualized metrics and logs with customizable CloudWatch dashboards.
- Used widgets to display graphs, tables, and maps.
- Set up CloudWatch Alarms based on metric conditions, log patterns, or by using Python programming with the Boto3 CloudWatch client: cloudwatch_logs = boto3.client('logs')
- Configure notification actions, such as sending emails or SNS messages.
- Developed and executed comprehensive migration plans, including application assessment, dependency mapping, and image optimization. Utilized Amazon CloudWatch Contributor Insights to Analyze Traffic Trends
- Proficient in designing and implementing robust observability solutions using CloudWatch, including custom metric creation and advanced log analysis with CloudWatch Insights. Leveraged Python (Boto3) for efficient metric publication.
- Demonstrated expertise in AWS infrastructure automation with CloudFormation and Python (Boto3) for efficient resource provisioning and management:
- Developed Mulesoft Anypoint APIs and Dashboards to provide real-time insights and monitoring of trading activities.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.

- Implemented automated security scans (SAST, DAST) into the GitLab CI pipeline, enforced granular access controls using roles, and managed secrets securely. Regular audits ensured compliance with industry standards like PCI-DSS or SOC 2.
- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
    - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **CI/CD for Integration Solutions:** Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (MuleSoft, Cloud Functions, or Hybrid).
- **Data Transformation Expertise:** Utilized MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations and DBT for standardization and basic transformations.
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
- **CI/CD and DevSecOps Automation:** Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.
- **AI-Powered Pattern Recognition:** Used Artificial Neural Networks to model complex patterns and forecast potential issues.
- **High-Performance Kubernetes Cluster:** Built a high-performance Kubernetes cluster on Nvidia DGX-1 systems with pre-installed DGX OS for running containerized AI, HPC, and industry-specific workloads.
- Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation & management of deployed ML models in production env.
- Delivered optimal ML solutions either using Python, R with SQL 2016 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Integration Solution Architect/Snr AWS DevOps Engineer /Java/NET/Python/Mulesoft/SP/K2 Developer
Responsibilities:

- Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications. Build and maintain all non-production and production environments.
- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- Designed and maintained robust CI/CD pipelines using platform-specific tools:
    1. **Azure: Azure DevOps**
    2. **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    3. **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- **Automated Kubernetes Deployments:** Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.
- Design and lead the migration of complex applications from Rancher/Docker on EC2 to ECS Fargate, optimizing performance and cost efficiency. Configured CloudWatch alarms to proactively monitor system health and performance.
- Built and managed ECS clusters and task definitions, ensuring seamless application deployment and management.
- Implemented robust CI/CD pipelines for automated builds, testing, and deployment to ECS Fargate.
- Established comprehensive monitoring and optimization strategies to maintain application health and performance.
- Successfully migrated critical applications with minimal downtime and disruption to business operations.
- **CloudWatch Observability Expertise:** Implemented comprehensive CloudWatch observability solutions, including metrics, logs, and dashboards. Leveraged CloudWatch Insights for advanced log analysis and troubleshooting.
- Optimized CloudWatch costs through efficient data retention and resource management.
- **Centralized management**: Use IAM roles and policies to manage access to CloudWatch resources.
- **Cost optimization**: Optimize CloudWatch costs by adjusting data retention periods and alarm configurations.
- **Data retention**: Determine appropriate retention periods for metrics and logs based on business requirements.
- **Data encryption**: Enable encryption for metrics and logs to protect sensitive information.

- **Regular review**: Regularly review dashboards, alarms, and logs to ensure they meet business needs.
- **CloudWatch Metric Filters**: Filter log data based on specific patterns to reduce noise.
- Design and implement secure & scalable CI/CD pipelines using AWS CodeBuild, RDS & S3 with Bitbucket repositories.
- Production) and managed resource allocation.
- **Secured API Access for Student Enrollment**: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of students' data.
- **CI/CD and DevSecOps Automation (AWS)**: Built and maintained CI/CD pipelines for various applications, employing DevSecOps practices using industry-standard tools (CodeBuild/GitHub) & following best practices for secure development
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or even on-premises environment.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer/Data Engineer /Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code (IaC) tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed Amazon Redshift with its built-in data processing capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- **Automation:** Created and maintained automated scripts using Python or Perl programming, along with Bash or Shell scripting, to streamline development and operational processes.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.
- Collaborate with development and operations teams to streamline the deployment process of our Python Web applications.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within AWS cloud architecture.
- **Security**: Secured secrets with HashiCorp Vault or AWS Secrets Manager and implemented network segmentation and firewalls. Integrated security scanning tools throughout development.
- Troubleshoot and resolve issues related to deployment, configuration, and performance.
- **Leverage AWS services for cloud-native design patterns**:
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- **Optimized CI/CD pipelines with DevOps tools**: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon Elastic Container Registry (ECR) and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.

- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: AWS DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using AWS API Gateway
- Deploy and maintain web applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.

- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Snr Back-end Python Engineer-Opus – Micheal – 28/7/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Snr Full Stack Developer(Java, C#.NET, Python/Django/React or Angular)/Software Engineer/Snr DevOps/Data Engineer/ /AI/ML/SP/D365 Developer /AI/ML/DL/BI/SharePoint/Dynamics 365 CRM Developer /DBA –Japan(7.5yrs) & Aust. (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Back-end Python Engineer/Snr DevOps/Data Engineer/Software Engineer /AI/ML/SP/D365 Developer /EDW
Responsibilities:

- Architected and Implemented **hybrid cloud infrastructure architectures** between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- Full Stack engineer skilled in React/TypeScript for frontend development, Python(Flask or Django) for backend development, in AWS cloud. Proficient in integration tools like MuleSoft and web frameworks (e.g., Flask/Django) to ensure seamless data flow using RESTful web services. Develop Python web application components using frameworks like Flask or Django.
- Skilled in building microservices applications using Python & JavaScript with frameworks like Flask & Node.js. Expertise in consuming RESTful web services to integrate with ACQSC's enterprise services layer, databases, & clouds hosted systems.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Leverage Docker for containerization, Docker Compose for local multi-container orchestration, and Kubernetes for large-scale container management. Deploy Docker Compose applications to cloud platforms like AWS using ECS Fargate, EKS, or AWS CodeDeploy. Collaborate with front-end developers to integrate user-facing elements with server-side logic.
- **Utilized powerful Geographic Information System (GIS) software**, such as **ArcGIS Desktop** or **QGIS**, for creating, editing, analyzing, and visualizing spatial data. Making photorealistic 3D maps in javascript using Cesium.js

- **Implemented and maintained Azure Maps and Bing 3D Maps**: Ensuring data accuracy, accessibility, and integration with enterprise systems. Built and leveraged geospatial visualizations to support development teams.
- Skilled in building microservices applications using Python & JavaScript with frameworks like Flask & Node.js. Expertise in consuming RESTful web services to integrate with ACQSC's enterprise services layer, databases, and host systems.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation templates or Terraform. Utilized AWS Control Tower for centralized management of multiple AWS accounts, enabling parallel processing for various projects.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
    - **Azure: Azure DevOps**
    - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama3.1, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Python or GoLang programming to design and implement **back-end** AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, RNN, ANN, GAN) based on dataset type, size, and desired result interpretability.
- Using Harrison AI platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Develop medical-tailored LLMs (BERT, GPT-4o, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- Bridge the gap between R and low-level C++ for precise control and data manipulation.
- Create a C++ file (e.g., `my_linear_function.cpp`) containing the C++ function for using in R with RStudio and Rccp package.
- Use the `sourceCpp()` function to compile and load the C++ code into R: sourceCpp("my_linear_function.cpp")

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Back-end Python Engineer/Snr DevOps/Data Engineer/Software Engineer /AI/ML/SP/D365 Developer
Responsibilities:

Built and secured hybrid cloud architecture connecting AWS VPCs and GCP Cloud VPCs: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.

Cloud-Agnostic Data Platform: Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.

Designed and developed Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/ J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Develop and maintain high-quality backend systems: Build, optimize, and secure Node.js/TypeScript applications, collaborating with frontend teams for seamless integration.

Utilized Google Maps Platform to provides photorealistic 3D tiles and integrated into the JavaScript application using the Maps JavaScript API and Cesium.js.

Leveraged powerful Geographic Information System (GIS) software, such as **ArcGIS Desktop** or **QGIS**, for creating, editing, analyzing, and visualizing spatial data. Making photorealistic 3D maps in javascript using Cesium.js

Developed and maintained data quality and reference data: Collaborated with teams to optimize geospatial data utilization.

Design, develop, and maintain a robust ETL pipeline/API orchestration using either:

1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics

Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.

Automation and CI/CD: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Early detection: Integrate security scanning tools (e.g., SonarQube, OWASP ZAP) into the CI pipeline, reducing the time and effort required for remediation.

Secure Coding Practices: Implement secure coding practices to minimize vulnerabilities in applications.

Vulnerability Management: Regularly conduct static code analysis and dynamic application security testing to identify and address potential vulnerabilities.

Secrets Management: Utilize secrets management tools (e.g., HashiCorp Vault) to securely store and manage sensitive credentials.

DevSecOps Security: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.

Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.

Champion DevOps innovation, leveraging agile methodologies and tech to streamline high-quality software delivery.

Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.

Leverage Docker for containerization, Docker Compose for local multi-container orchestration, and Kubernetes for large-scale container management. Deploy Docker Compose applications to cloud platforms like AWS using ECS Fargate, EKS, or AWS CodeDeploy. Collaborate with front-end developers to integrate user-facing elements with server-side logic.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Crafted **back-end** AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, using SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.

Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.

Developed scalable web applications in Java, C#.NET, Python/Django/React or Angular, & deployed them on Amazon Relational Database Service (RDS), ensuring seamless integration with DevOps workflow using tools like Jenkins, GitLab CI/CD, or AWS CodePipeline.

Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu + Nvida tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.

Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Scalability and Optimization: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Back-end Python Engineer/Snr DevOps/Data Engineer/Software Engineer /AI/ML/SP/K2 Developer
Responsibilities:

Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.

Develop & maintain high-quality backend systems for web applications using Python(Flask or Django), Node.js &TypeScript.

Frequently implemented AWS cloud-native design patterns. Such as **Microservices Architecture** to break down a large application into smaller, independent services. **Serverless Computing** to run code without provisioning or managing servers, **API Gateway** to manage, secure, and monitor the APIs in a central location.

Used Terraform & Helm to automate deployments of Amazon Elastic Kubernetes Service (EKS) clusters for microservices.

Designed and implemented CI/CD pipelines across AWS CodePipeline, GitHub Actions, and Jenkins, utilizing scripting languages (Python, Ansible, etc.) for automated tasks, to ensure continuous delivery of applications.

Utilized Amazon CloudWatch, logging, partitioning, indexing, & optimizing AWS Glue pipeline structure to analyze system & application performance, providing insights for business case development to enhance service delivery and reduce issues.

Secured AWS resources using AWS Identity and Access Management (IAM), AWS Key Management Service (KMS), AWS Config, & AWS Security Hub. Monitored and troubleshooted AWS services using Splunk, Data Dog and AWS CloudWatch.

Docker Images: Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.

Successfully migrated on-premises workloads to AWS Cloud, including data from IBM TM1 OLAP DB, PostgreSQL, and Oracle databases to Amazon RDS. This involved leveraging AWS Glue Data Catalog for initial data access, followed by the AWS Database Migration Service for comprehensive database migration. Additionally, AWS Migration Hub and AWS Application

Migration Service ensured seamless infrastructure and application migration, while the Power BI SQL Server Data Connector facilitated smooth data connectivity from migrated SQL Server databases.

Crafted **back-end** AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, using SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

API Development with Python: Developed ML models using Python frameworks (e.g., Flask, FastAPI) and secured RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.

Developed scalable web applications in Java, C#.NET, Python/Django/React or Angular, & deployed them on Amazon RDS, ensuring seamless integration with DevOps workflow using tools like Jenkins, GitLab CI/CD, or AWS CodePipeline.

Leverage Amazon Kinesis, AWS Lambda, and Amazon API Gateway for AI/ML, IoT, AR/VR Development: Developed and deployed cloud-native applications using AWS Lambda, AWS ECS, and AWS SQS.

Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM2016/Power BI/Power Automate etc.)

K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.

Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Leveraged DataRobot's automated ML platform and best practices (Agile, DevOps, Configuration-As-Code) to develop and deploy end-to-end machine learning solutions for clients.

Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Full Stack Developer/Software Test Engineer/Cloud/Linux Administrator/DBA

Responsibilities:

- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Generated test plans, status reports, and summaries to keep stakeholders informed.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows
- Design and develop cross-platform (Andriod/iOS) mobile Apps either using Xamarin or React Native with Javascript and React Library or Nativescript with Angular2 without DOM, Cross-Compilation and Plugins in Visual Studio Code.
- Developed and deployed cloud-native applications using AWS Lambda, AWS ECS, and AWS SQS.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- **Secure Secrets**: Use tools like HashiCorp Vault or AWS Secrets Manager to safeguard sensitive information.
- **Patch Regularly**: Keep systems and applications up to date with security patches.
- **Network Segmentation**: Implement network segmentation and firewalls.
- **Compliance and Scanning**: Ensure compliance with industry standards (e.g., GDPR, HIPAA) and integrate security scanning tools (e.g., SonarQube, OWASP ZAP) into your development pipeline to detect and address vulnerabilities early
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.

Proficiently set up and managed AWS environments, including the establishment and management of AWS secure landing zones via AWS Landing Zone; Create EC2 instances either by using AWS CLI or AWS SDK.

Successfully installed, configured, and tested new and existing hardware, software, and WAN/LAN technologies.

- Defined, developed, and documented automation technologies to improve system efficiency and streamline build, test, and deployment processes. Collaborated with development teams to implement automated build and deployment pipelines using AWS CodePipeline, YAML, CloudFormation, and AWS CLI.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.

- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Fully Stack Developer/DevOps Engineer/Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- Developed scalable web applications in Java, C#.NET, Python/Django/React or Angular, & deployed them on Azure SQL Database, ensuring seamless integration with DevOps workflow using tools like Jenkins, GitLab CI/CD, or Azure DevOps.
- Automation Expertise: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Actively contribute to and guide innovation in DevOps practices and incorporate agile delivery approaches and technologies that assist with the development and delivery of high-quality software
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Data Engineer/Senior Systems Engineer /DBA
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using API Gateway
- Deploy and maintain web applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR1, and Access Manager for Oracle-deployed applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.

- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbor's algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Gemini Pro 1.5, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

AI Engineer – KPMG – 11/7/24
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
AI Engineer/Software Engineer/Full Stack Developer(Java/.NET/Python) /Senior DevOps Engineer /Snr Data Engineer / Integration Solution Architect /Salesforce/Mulesoft/AI/ML/DL/BI/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: AI Engineer /Software Engineer/Snr DevOps/Data Engineer/Python/AI/ML/DL/SP/D365 CRM Developer /EDW
Responsibilities:

- Designed and implemented scalable AI solutions in hybrid cloud platforms (AWS SageMaker, Azure ML, Azure OpenAI Service and GCP Vertex AI) using modular and microservices architectures for efficient development and deployment.

- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation templates or Terraform. Utilized AWS Control Tower for centralized management of multiple AWS accounts, enabling parallel processing for various projects.In my role as a key AI Engineer, I spearhead the development and deployment of impactful AI solutions across the Commission, driving innovation as part of a dynamic team.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro.
- Develop and deploy LLMs efficiently with LangChain. Perform document analysis and summarization on PDFs using pdfread. Streamline LLM application development with PrompFlow. Enable context awareness, decision-making, and task automation in Multi-GPTs and Auto-Agents with a semantic kernel.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Deigned and implemented data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.

- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Designed and maintained robust AI/ML pipelines using platform-specific tools:
  - Azure: Azure Machine Learning (AML), Azure OpenAI Service
  - AWS: AWS SageMaker
  - GCP: GCP Vertex AI
- Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- Leveraged automation to accelerate the creation of brain organoids for neurological disease research.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
  - **Azure: Azure DevOps**
  - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
  - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Full-stack engineer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, and AWS cloud. Proficient in integration tools like MuleSoft and web frameworks (e.g., Flask/Django) to ensure seamless data flow using RESTful web services. Develop Python web application components using frameworks like Flask or Django.
- Skilled in building microservices applications using Python & JavaScript with frameworks like Flask & Node.js. Expertise in consuming RESTful web services to integrate with ACQSC's enterprise services layer, databases, and host systems.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: AI Engineer /Senior DevOps Engineer/Mulesoft/Java/Python/R/AI/ML/DL/BI/SharePoint Developer/EDW
Responsibilities:

Hybrid Cloud Architect: Established secure connections between AWS, GCP, and Azure cloud environments for data transfer. Utilized infrastructure-as-code tools (Terraform/Ansible, or Puppet) for automated and consistent provisioning.

Built secure and scalable GCP Landing Zones using IaC with Terraform or Cloud Build. Utilized GCP Cloud IAM for centralized governance & management of multiple GCP projects, enabling efficient parallel processing for various projects.

Developing and implementing AI solutions, translating business requirements into viable technical solutions.

Collaborating closely with stakeholders to ensure alignment between business objectives and AI capabilities.

Delivered robust AI models and infrastructure through development, maintenance, and clear communication of design principles and strategic impact to stakeholders.
Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Developed Mulesoft Anypoint APIs and Dashboards to provide real-time insights and monitoring of trading activities.
Full-stack engineer skilled in Angular/TypeScript for frontend development, Python/Flask or Django for backend development, and AWS cloud. Proficient in integration tools like MuleSoft and web frameworks (e.g., Flask/Django) to ensure seamless data flow using RESTful web services.
Develop Python web application components using frameworks like Flask or Django.
Skilled in building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with RBA's enterprise services layer, databases, and host systems.
Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
Implemented automated security scans (SAST, DAST) into the GitLab CI pipeline, enforced granular access controls using roles, and managed secrets securely. Regular audits ensured compliance with industry standards like PCI-DSS or SOC 2.

- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
  - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
  - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
Automation and CI/CD: Leveraged industry-standard tools like Helm and Kubespray to streamline & customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
Streamlined Terraform configuration management through the strategic use of the merge function to consolidate settings from diverse sources.
Cloud-Agnostic Data Platform: Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
CI/CD for Integration Solutions: Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (MuleSoft, Cloud Functions, or Hybrid).
Data Transformation Expertise: Utilized MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations and DBT for standardization and basic transformations.
Implement effective branching strategies utilizing Gitflow for seamless deployments and continuous application monitoring.
Optimized CI/CD pipelines with DevOps tools: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
Production-Ready Machine Learning Models: Engineered and deployed production-ready ML models for real-world impact using Python and MLOps tools.
Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
CI/CD and DevSecOps Automation: Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.
AI-Powered Pattern Recognition: Used Artificial Neural Networks to model complex patterns and forecast potential issues.
High-Performance Kubernetes Cluster: Built a high-performance Kubernetes cluster on Nvidia DGX-1 systems with pre-installed DGX OS for running containerized AI, HPC, and industry-specific workloads.
Delivered optimal ML solutions either using Python, Scala, R with SQL 2016 ML feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow & scikit-learn on AWS.
Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: AI Engineer/Integration Solution Architect/Snr DevOps/Data Engineer/Java/Mulesoft/SP/K2 Developer
Responsibilities:
Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications. Build and maintain all non-production and production environments.
Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.

Designed and maintained robust CI/CD pipelines using platform-specific tools:

4. **Azure: Azure DevOps**
5. **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
6. **GCP: Cloud Build/Cloud Deploy**

Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:

Python with Pandas and Airflow to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or

Building microservices applications using Python & JavaScript with frameworks like Flask & Node.js. Expertise in consuming RESTful web services to integrate with UC's enterprise services layer, databases, and host systems.

Automated Kubernetes Deployments: Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.

Developed and deployed secure, RESTful Python APIs for web applications using frameworks like Flask or FastAPI. These APIs can integrate with MLOps pipelines for data access, model predictions, or other machine learning tasks.

Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.

Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics CRM 2016.

Configure Connections: Provide access credentials, URLs, and other details for each system within the connectors.

Design Integration Logic: Transform data, handle errors, and implement security measures within the Mule flow.

Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.

Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.

Design and implement secure & scalable CI/CD pipelines using AWS CodeBuild, RDS & S3 with Bitbucket repositories.

Integration Solution Design with Azure DevOps CI/CD: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).

Develop and Build: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).

Deployment Stages: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).

Developed and deployed a MuleSoft application: Used Anypoint Studio to design, build, and package the application. Configured deployment settings for different environments (Dev, UAT, Production) and managed resource allocation.

Secured API Access for Student Enrollment: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of students' data.

CI/CD and DevSecOps Automation (AWS): Built and maintained CI/CD pipelines for various applications, employing DevSecOps practices using industry-standard tools (CodeBuild/GitHub) & following best practices for secure development

- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or even on-premises environment.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation & management of deployed ML models in production env.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**

Healthcare Sector: DevOps Engineer/Data Engineer /Software Test Engineer/Cloud/Linux Administrator

Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed Amazon Redshift with its built-in data processing capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.

- **Automation:** Created and maintained automated scripts using Python or Perl programming, along with Bash or Shell scripting, to streamline development and operational processes.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.
- Collaborate with development and operations teams to streamline the deployment process of our Python Web applications.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within AWS cloud architecture.
- **Security**: Secured secrets with HashiCorp Vault or AWS Secrets Manager and implemented network segmentation and firewalls. Integrated security scanning tools throughout development.
- Troubleshoot and resolve issues related to deployment, configuration, and performance.
- **Leverage AWS services for cloud-native design patterns**:
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon Elastic Container Registry (ECR) and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using API Gateway
- Deploy and maintain web applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.